

AI Engineer

Step 2: Master Machine Learning

CareerCompass • Detailed Concept Notes • Research-backed learning guide

Concept & Learning Objective

Machine learning is a disciplined workflow, not simply calling a model API. Start with the problem definition and data, establish a baseline, split data correctly, train, evaluate, analyze errors, tune and monitor.

Scikit-learn's user guide provides practical coverage of supervised learning, preprocessing, model selection, evaluation and pipelines. The central engineering concern is preventing leakage and ensuring that the same transformations used during training are applied during inference.

Core Concepts — Detailed Breakdown

- Problem framing: target, features, constraints and success metric.
- Supervised vs unsupervised learning.
- Regression vs classification.
- Data splitting: train/validation/test and time-aware splitting when appropriate.
- Preprocessing: scaling, encoding, missing values and feature engineering.
- Algorithms: linear/logistic regression, trees, random forests, boosting, nearest neighbors and clustering.
- Overfitting/underfitting and bias/variance.
- Metrics: accuracy, precision, recall, F1, ROC-AUC, MAE, MSE/RMSE and task-specific metrics.
- Pipelines: combine preprocessing and estimator to prevent inconsistent transformations.
- Cross-validation and hyperparameter search.

What You Should Be Able To Do

- Build a baseline before a complex model.
- Choose metrics based on the cost of errors.
- Use a pipeline for preprocessing.
- Compare at least three models.
- Perform error analysis instead of relying only on one score.

Hands-on Project / Practice

- Build a classification project with a reproducible preprocessing pipeline. Report baseline, cross-validation results, final test result and examples of false positives/false negatives.

Recommended References

- Scikit-learn User Guide — https://scikit-learn.org/stable/user_guide.html
- Kaggle — <https://www.kaggle.com/>
- Machine Learning with Python and Scikit-Learn — <https://www.youtube.com/watch?v=hDKCxebp88A>

Note: These notes are designed to teach the concepts and provide a practical learning path. Always prefer the linked official documentation for current APIs, SDK versions and platform-specific release requirements.